HAT-P-36b

R-0667

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_260527` · created 2026-07-27T18:11:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461187.8116 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.012
Transit depth (Rp/Rs)^2	2.35 +/- 0.38 [%]
Orbital Inclination (inc)	82.84 +/- 2.18
Airmass coefficient 1 (a1)	1.455 +/- 0.023
Airmass coefficient 2 (a2)	-0.1671 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	1.58 %
Best Comparison Star	None
Optimal Aperture	3.54
Optimal Annulus	13.27
Transit Duration (day)	0.0867 +/- 0.0091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.012 vs 0.126 (deviation 1.97σ)
Duration fit vs published	0.0867 d vs 0.0925 d (deviation 0.56σ)
Mid-time O–C	-16.6 min
Depth SNR	11.2
Residual scatter	1.58 %

Light curve

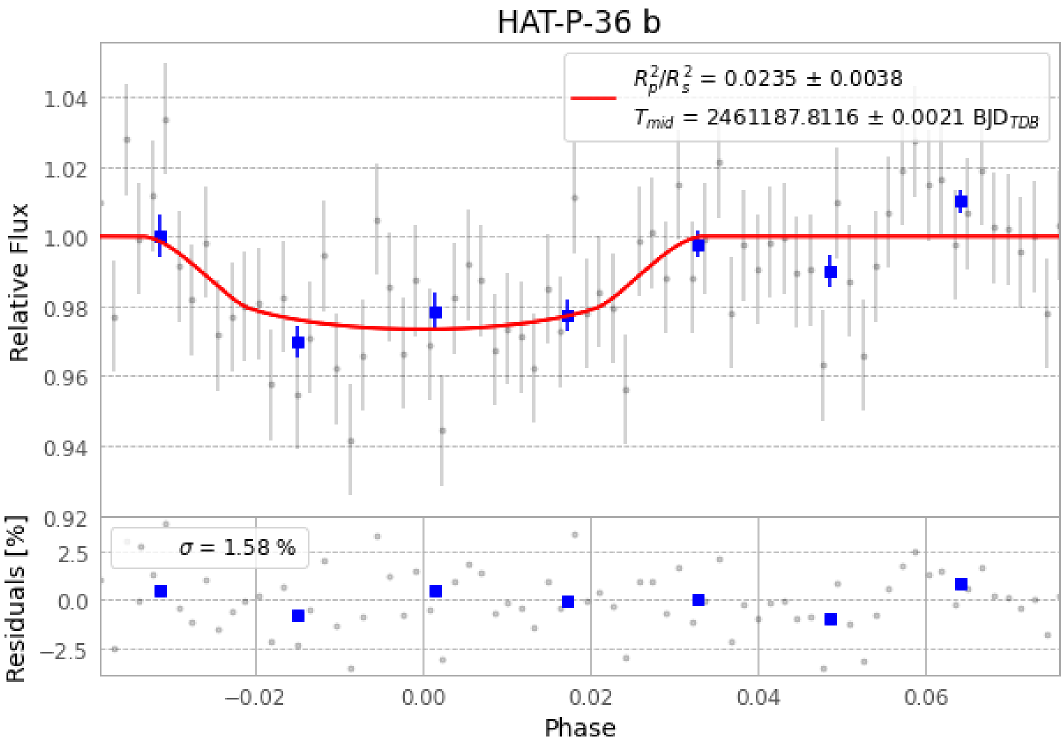

Figure 1 — FinalLightCurve_HAT-P-36 b_2026-05-26.png (SHA-256 72d732435bba2dcc…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [257, 207], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 61ee9538fed626aa…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d476b1b`

Data provenance

74 FITS frames (49 MB), `HATP-36260527061215.FITS` → `HATP-36260527095115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260527.log (5c2cf3bc56b4…), FinalLightCurve_HAT-P-36 b_2026-05-26.png (72d732435bba…), FinalLightCurve_HAT-P-36 b_2026-05-26.csv (a3fb46ea2fdd…)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-27 18:11Z.